Koa Shen

MANUFACTURING ENGINEERING | SUMMER 2027

(415) 810-8344 | koashensf@gmail.com | linkedin.com/in/koa-shen | koashen.com

PROFESSIONAL SUMMARY

Mechanical engineering student with hands-on CNC and manual machining, CAM, DFM, GD&T, precision inspection, and production-fixture experience. Combines shop-floor problem solving with CAD, process documentation, and engineering analysis.

EDUCATION

University of California, Santa Barbara | B.S. Mechanical Engineering
Santa Barbara, CA | Expected June 2028 | Major GPA: 3.97 | Cumulative GPA: 3.94
Dean's Honors every quarter | Relevant coursework: Mechanics of Materials, Dynamics, Circuits, MATLAB

MANUFACTURING AND ENGINEERING EXPERIENCE

Fabrication Team Lead | Gaucho Racing, Formula SAE Electric, UC Santa Barbara
Fall 2024–Present | Santa Barbara, CA

- CNC-machined 33 suspension components from 4130 steel and 7075-T6 aluminum on Haas mills and lathes; held clevis and rod-end bores within a 0.0004-in tolerance window and bearing interfaces within 0.0005 in, verified with go/no-go tests.
- Worked with designers on GD&T and DFM; proposed a standardized spindle interface to reuse CAM programs and simplify setup and programming.
- Led 15 teammates through driver-safety-system design, fabrication, and installation; GR26 passed Formula SAE technical inspection.

Mechanical Engineering Intern | SLAC National Accelerator Laboratory, LCLS
Summer 2026–Present | Menlo Park, CA; extended part-time remote

- Created Siemens Solid Edge designs and Teamcenter PDM-controlled documentation with GD&T for manufacture and inspection of beamline motion assemblies.
- Built a Python/Drake digital twin for a 30-DOF, 305-body assembly; millisecond collision checks identified at least five pinch points and informed design changes.
- Developed 15 reusable motion-stage models; the framework is being adopted across three LCLS assemblies.

Maintenance Technician | Exploratorium
Summer 2025 | San Francisco, CA

- Serviced 100+ public exhibits; reverse-engineered components and developed CAD, CAM, and fixturing for replacement parts using manual machining and welding.
- Redesigned a load-spreading bezel to address acrylic cracking and avoid $3,000+ in replacement costs; documented remanufacturing instructions for future technicians.

SELECTED MANUFACTURING PROJECT

FSAE Chassis Welding Fixtures | Gaucho Racing

- Designed and built locating fixtures for chassis tubes; GR25 fixture positioned 81 tubes within 0.050 in of CAD node locations and cut assembly time 50% versus the prior fixture.
- Developed a GR26 hybrid fixture using 3D-printed locating brackets and reused 80/20 and plywood, improving assembly and reducing fixture assembly time a further 30%.

TECHNICAL SKILLS

Manufacturing: CNC milling and turning, Haas machines, Mastercam, CAM, manual machining, welding, fixturing, 3D printing, assembly
Quality and Design: GD&T, DFM, go/no-go inspection, precision measurement, Solid Edge, SolidWorks, Teamcenter PDM, FEA
Engineering Software: Python, C/C++, MATLAB, Drake, Git